

Entrainment and Asymmetry in Spatially Developing Supersonic Turbulent Shear Layers

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Direct Numerical Simulations (DNS) of a spatially developing supersonic turbulent shear layer have been conducted at different convective Mach numbers and velocity parameters to examine the self-similarity, transverse velocity, flow asymmetry, and entrainment of the layer. Self-similarity is achieved at far enough downstream locations for all the simulated cases. The self-similar normalized density distribution inside the shear layer and the centerline shifting of the layer toward the low-speed stream are used to explain the asymmetry in the far-field transverse velocities on the high-speed and low-speed sides. An approximate equation for the transverse velocity is developed and the profiles of transverse velocity obtained from this equation are compared with the DNS results. The centerline shifting and the approximate equation for transverse velocity are used to obtain a general expression for the entrainment ratio of the two streams. The entrainment ratio, favoring excess entrainment on the high-speed side, is shown to increase with the convective Mach numbers and velocity parameters.

I. Nomenclature

a	=	speed of sound
δ	=	shear layer thickness
$\bar{x}_2$	=	centerline shifting
M	=	mach number
U_1	=	primary freestream velocity
U_2	=	secondary freestream velocity
ΔU	=	freestream velocity difference
M_c	=	convective Mach number = $\frac{U_1 - U_2}{a_1 + a_2}$
λ	=	velocity parameter = $\frac{U_1 - U_2}{U_1 + U_2}$
r	=	velocity ratio = $\frac{U_2}{U_1}$
ρ	=	density
γ	=	specific heat ratio = $\frac{C_p}{C_v}$
U_c	=	average velocity = $\frac{U_1 + U_2}{2}$

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II. Introduction

THE study of the mixing layers has received considerable attention in the last several decades due to their complex flow physics and a wide range of engineering applications [1–13]. Mixing layers are encountered in air-breathing propulsion systems [7], supersonic ejectors [14], missiles and projectile flows [15], noise generating engine jets [8], the slipstream behind wings and the interface between various streams in a recirculation region [16].

This paper is focused on supersonic compressible shear layers. The fundamental understanding of supersonic/compressible shear layers is essential to overcome challenges associated with modified flow dynamics, reduced growth rate, changed mixing, and chemical reactions. Incompressible shear layers have been extensively studied but there are only limited studies of compressible supersonic shear layers. The reduced growth rate with compressibility is well known and makes the application challenging [7]. However, much less is known about transverse velocity, entrainment, and asymmetry. A consistent self-similar description of the compressible shear layers is also missing, so are the effects of important flow parameters on the self-similar flow variables. The parameter that quantifies the compressibility of the shear layer is the convective Mach number ($M_c = \frac{U_1 - U_2}{a_1 + a_2}$) and was considered in several past studies [6, 7, 12, 16–18]. Additionally, there is another important parameter that affects the growth rate and statistical behavior of the shear layer, called the velocity parameter ($\lambda = \frac{U_1 - U_2}{U_1 + U_2}$) [9], that is important in both incompressible and compressible shear layers. For shear layers with constant global density, these are the two most important parameters affecting the flow [19].

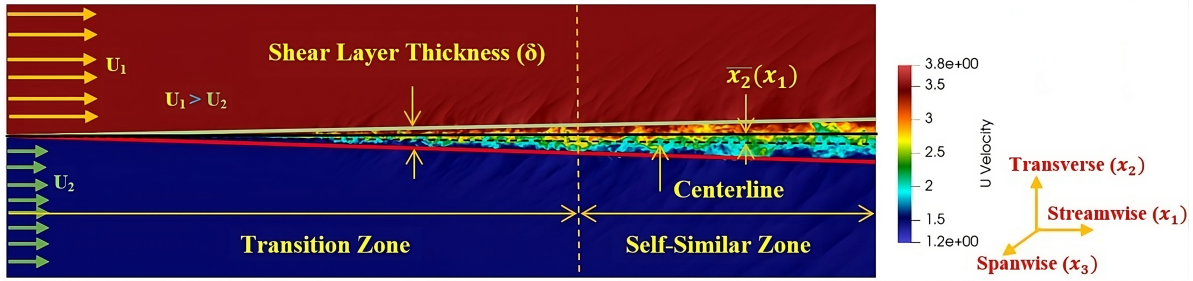


Fig. 1 Schematic of spatially developing three-dimensional shear layer. The primary (high-speed) and secondary (low-speed) free streams are represented by subscripts 1 and 2, respectively. (—) line represents the upper edge and (—) line represents the lower edge of the shear layer. (—) is the geometric center of the shear layer. (---) line is the shifted centerline and $\bar{x}_2$ is the shifting distance from the geometric centerline. The average densities of the two streams are the same in this study. The coordinate system is chosen in such a way that x_1 represents the streamwise direction, x_2 represents the transverse or cross-stream direction, and x_3 represents the spanwise direction.

While experimental studies of shear layers naturally involve the spatially developing features like asymmetry, DNS studies of compressible shear layers are mostly restricted to the temporal case and cannot capture these features. Furthermore, a "complete" self-similar analysis of spatially developing compressible shear layers is missing in the literature. There is also limited information on the behavior of the transverse velocity profile, asymmetry, and entrainment of the spatial shear layer. To address these shortcomings, Direct Numerical Simulations (DNS) of a spatially developing supersonic turbulent shear layer are conducted for a range of convective Mach numbers (M_c) and velocity parameters (λ) to examine the combined effects of compressibility and advection on self-similarity, transverse velocity, asymmetry, and entrainment of the layer. In this paper, we consider compressibility and centerline shifting in the self-similar analysis. We show that the convective Mach number and velocity parameter appear naturally in the self-similar equations. Then, we derive the approximate equation for the transverse velocity and compare the profiles obtained from this equation with our DNS results. The asymmetric behavior of transverse velocity is linked to the centerline shifting and density variation in the core of the shear layer. Utilizing the transverse velocity and centerline shifting, we provide the general expression for the entrainment ratio and we show how it varies with compressibility and velocity ratio.

III. Computational Approach

In this study, Direct Numerical Simulations have been conducted to simulate the spatially developing supersonic turbulent shear layer. The Reynolds number, based on the inflow vorticity thickness, is kept at 640. The molecular Prandtl and Schmidt numbers are $Pr = Sc = 0.72$. The supersonic inflow is specified by a tangent hyperbolic profile for

the mean streamwise velocity as:

$$\bar{u}_1(x_2) = \frac{U_1 + U_2}{2} + \frac{U_1 - U_2}{2} \tanh\left(\frac{2x_2}{\delta_{w,0}}\right) \quad (1)$$

where the initial vorticity thickness $\delta_{w,0}$ represents the initial thickness of the interface, $\bar{u}_2=0$ and $\bar{u}_3=0$. The hyperbolic tangent profile stands out as the most commonly utilized function in simulations of shear layers [8, 10, 11, 20–23]. Uniform temperature and density are maintained at the inlet, with a constant pressure of $1/\gamma M^2$. To induce instability, random inflow perturbations are introduced and added to mean inlet velocities. The scalar profile (not discussed in this paper) follows a similar trend to the mean streamwise velocity having a value of 1 at the upper free stream and 0 at the lower free stream. A supersonic outflow boundary condition is employed at the outlet, while periodic boundary conditions are applied in the spanwise directions. To prevent wave reflections back into the computational domain, non-reflective boundary conditions are employed at the upper and lower boundaries [24].

In this study, the sixth-order compact scheme is utilized to evaluate the viscous fluxes [24], and the seventh-order MP scheme is employed to compute the inviscid fluxes [11]. The 3rd-order Runge-Kutta scheme is used for the time advancement [25]. Convergence tests are conducted to establish the accuracy of numerical results for a wide range of grid and domain sizes. Domain size is large enough in the streamwise and transverse directions for self-similar growth to be established and boundary conditions effects on flow dynamics to be made negligible. Two distinct sets of simulations are conducted. In the first set of simulations, five different convective Mach numbers ($M_c = 0.8, 1.0, 1.2, 1.4, 1.6$) are chosen, keeping the velocity parameter constant ($\lambda=0.4$) and in the second set of simulations, four different velocity parameters ($\lambda=0.2, 0.3, 0.4, 0.5$) are chosen, keeping the convective Mach number constant ($M_c=1.2$).

IV. Self-similarity

It is expected that the shear layer will exhibit self-similar growth after the initial transition. Shear layers can be studied using Prandtl's boundary layer equations since they are thin and spread slowly in the cross-stream directions [9]. Assuming an ideal gas behavior, steady-state conditions, and negligible variations in the spanwise direction, the compressible mean continuity and x_1 momentum after applying the boundary layer approximation can be expressed as:

$$\frac{\partial(\bar{\rho}\bar{u}_1)}{\partial x_1} + \frac{\partial(\bar{\rho}\bar{u}_2)}{\partial x_2} = 0. \quad (2)$$

$$\bar{\rho}\bar{u}_1 \frac{\partial \bar{u}_1}{\partial x_1} + \bar{\rho}\bar{u}_2 \frac{\partial \bar{u}_1}{\partial x_2} + \frac{\partial(\bar{\rho}\bar{R}_{12})}{\partial x_2} = 0. \quad (3)$$

$\bar{u}_1$ and $\bar{u}_2$ are the Favre average mean velocity in the streamwise and transverse directions respectively. The Favre averaged kinematic Reynolds stress is defined as $\bar{R}_{ij} = \frac{\langle \rho u_i u_j \rangle}{\langle \rho \rangle}$.

The statistics of a spatially developing shear layer depend on the streamwise x_1 and inhomogeneous transverse x_2 directions. Therefore, by introducing a new variable η , the location-wise dependencies can be eliminated. η can be defined as:

$$\eta = \frac{x_2 - \bar{x}_2}{\delta(x_1)}, \quad (4)$$

where δ is considered as the shear layer thickness, and $\bar{x}_2$ is the centerline shifting distance as shown in figure 1. $\bar{x}_2$ is the location where the streamwise velocity becomes equal to the average velocity. The mean continuity and momentum equation are self-similar if the growth rate $\frac{d\delta}{dx_1}$ and the centerline shifting speed $\frac{d\bar{x}_2}{dx_1}$ are constant and the variables are non-dimensionalized as:

$$\bar{\rho}(x_1, x_2) = \rho_{01} \hat{\rho}(\eta) + \rho_0 \quad (5)$$

$$\bar{P}(x_1, x_2) = P_{01} \hat{P}(\eta) + P_0 \quad (6)$$

$$\bar{u}_1(x_1, x_2) = u_{01} \hat{u}_1(\eta) + U_c \quad (7)$$

$$\bar{u}_2(x_1, x_2) = u_{02} \hat{u}_2(\eta) + u_2^* \quad (8)$$

$$\bar{R}_{11}(x_1, x_2) = R_{011} \hat{R}_{11}(\eta) \quad (9)$$

$$\bar{R}_{12}(x_1, x_2) = R_{012} \hat{R}_{12}(\eta) \quad (10)$$

ρ_{01} , P_{01} , u_{01} , u_{02} , R_{011} , and R_{012} are the reference scales of the mean density, pressure, streamwise velocity, transverse velocity, Reynolds normal stress in the streamwise direction, and Reynolds shear stress respectively. u_2^* is the mean

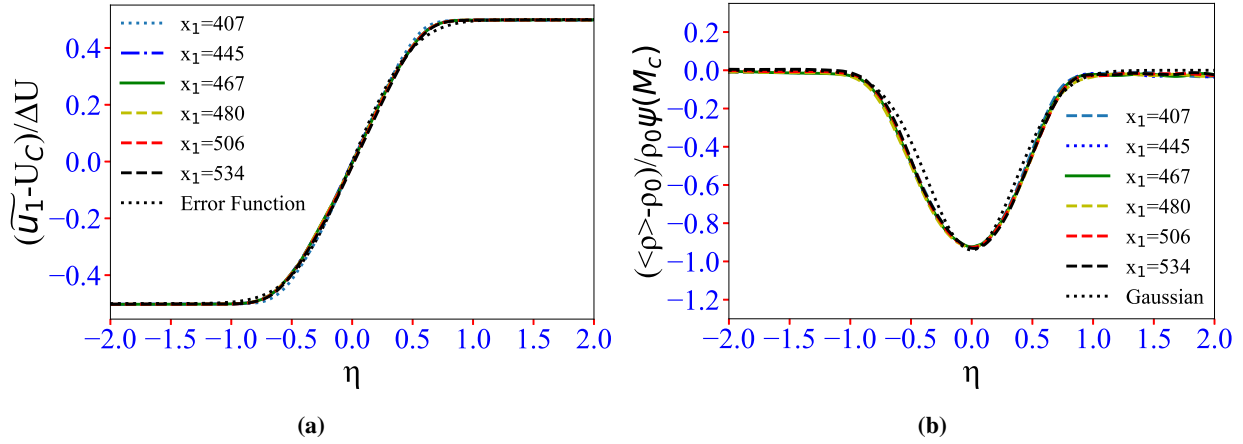


Fig. 2 Collapse of (a) scaled streamwise velocity and (b) scaled density in the self-similar coordinate. The scaled mean streamwise velocity can be approximated as an Error function ($\hat{u}_1 = A \text{erf}[B\eta]$) [9, 31] and the scaled density can be approximated as a Gaussian function ($\hat{\rho} = a e^{-B^2 \eta^2}$).

transverse velocity at the shear layer centerline ($\eta=0$). ρ_0 , and P_0 are the initial density, and pressure, respectively. U_c is the convective or average velocity of the two streams. These reference scales are obtained by the scaling patch approach [9, 26, 27]. Substituting the self-similar variables and their derivatives into equation 2, the mean continuity equation can be expressed as follows:

$$-(\eta + \phi) \left[\psi(M_c) \frac{d(\hat{\rho}\hat{u}_1)}{d\eta} + \frac{\psi(M_c)}{2\lambda} \frac{d\hat{\rho}}{d\eta} + \frac{d\hat{u}_1}{d\eta} \right] + \left[\psi(M_c) \frac{u_{02}}{u_{01}} \frac{d(\hat{\rho}\hat{u}_2)}{d\eta} + \psi(M_c) \frac{u_2^*}{u_{01}} \frac{d\hat{\rho}}{d\eta} + \frac{u_{02}}{u_{01}} \frac{d\hat{u}_2}{d\eta} \right] = 0. \quad (11)$$

$$\text{where } \phi = \frac{\frac{d\tilde{x}_2}{dx_1}}{\frac{d\delta}{dx_1}}. \quad (12)$$

ϕ is the ratio between the centerline shifting speed and the growth rate of the shear layer [9]. To make equation (11) self-similar, it is required that ϕ is a constant. Present DNS data suggest that in the far downstream region, both $\frac{d\tilde{x}_2}{dx_1}$ and $\frac{d\delta}{dx_1}$ are constant, which makes ϕ constant. In the present coordinate system, ϕ is negative because $\frac{d\tilde{x}_2}{dx_1}$ is negative. Experimental data also suggests that ϕ is negative and constant [9, 28–30]. To make the equation (11) self-similar, $\tilde{u}_2$ has to be scaled with $u_{01} \frac{d\delta}{dx_1}$. If the scaling of mean streamwise velocity is ΔU [9, 10], then $\Delta U \frac{d\delta}{dx_1}$ is the self-similar scaling for $\tilde{u}_2$. Furthermore, according to the definition of velocity parameter, $\frac{U_c}{u_{01}}$ is written as $\frac{1}{2\lambda}$ in the equation (11). There is a significant density variation inside the core of the shear layer due to viscous dissipation and the drop in density increases with increasing compressibility. To collapse density profiles at different convective Mach numbers, ρ_{01} is written as $\rho_0 \psi(M_c)$. $\psi(M_c)$ can be any function of M_c which satisfies the behavior of $\hat{\rho}$ at both limiting conditions (i.e., M_c goes to 0 or M_c goes to ∞) and collapses the data. Here, we are suggesting a Gaussian function of M_c ($\psi(M_c) = 1 - e^{-0.17 M_c^2}$) which collapses our data. However, any function that collapses the data and satisfies the behavior of $\hat{\rho}$ at both limiting conditions can be used as $\psi(M_c)$.

In a self-similar flow, the viscous terms have little effect on mean momentum. After applying the self-similar variables and their derivatives to the mean x_1 momentum equation, the equation becomes:

$$-(\eta + \phi) \left[2\lambda \psi(M_c) \hat{\rho} \hat{u}_1 \frac{d\hat{u}_1}{d\eta} + \psi(M_c) \hat{\rho} \frac{d\hat{u}_1}{d\eta} + 2\lambda \hat{u}_1 \frac{d\hat{u}_1}{d\eta} + \frac{d\hat{u}_1}{d\eta} \right] + \left[2\lambda \psi(M_c) \frac{u_{02}}{u_{01}} \hat{\rho} \hat{u}_2 \frac{d\hat{u}_1}{d\eta} + 2\lambda \psi(M_c) \frac{u_2^*}{u_{01}} \hat{\rho} \frac{d\hat{u}_1}{d\eta} + 2\lambda \frac{u_{02}}{u_{01}} \hat{u}_2 \frac{d\hat{u}_1}{d\eta} + 2\lambda \frac{u_2^*}{u_{01}} \frac{d\hat{u}_1}{d\eta} \right] + \psi(M_c) \frac{R_{012}}{u_{01}^2} \frac{d(\hat{\rho}\hat{R}_{12})}{d\eta} + \frac{R_{012}}{2\lambda} \frac{d(\hat{R}_{12})}{d\eta} = 0. \quad (13)$$

Since the scaling of $\tilde{u}_1$ is ΔU and of $\tilde{u}_2$ is $\Delta U \frac{d\delta}{dx_1}$ as discussed earlier, the self-similar x_1 momentum equation indicates that the scaling for $\hat{R}_{12}$ is $[\frac{1}{2\lambda} \frac{d\delta}{dx_1}] (\Delta U)^2$ or $\Delta U U_c \frac{d\delta}{dx_1}$. Furthermore, the self-similar scaling for $\hat{R}_{11}$ is $(\Delta U)^2$ [9, 10].

Utilizing the self-similar scalings, the ultimate self-similar expressions for the continuity and x_1 momentum equations are as follows:

$$-(\eta + \phi) \left[2\lambda \frac{d\hat{u}_1}{d\eta} (1 + \psi(M_c) \hat{\rho}) + \psi(M_c) \frac{d\hat{\rho}}{d\eta} (1 + 2\lambda \hat{u}_1) \right] + 2\lambda (1 + \psi(M_c) \hat{\rho}) \frac{d\hat{u}_2}{d\eta} + 2\lambda \psi(M_c) (\hat{u}_2 + \frac{u_2^*}{u_{02}}) \frac{d\hat{\rho}}{d\eta} = 0 \quad (14)$$

$$-(\eta + \phi)[(1 + 2\lambda\hat{u}_1)(1 + \psi(M_c)\hat{\rho})]\frac{d\hat{u}_1}{d\eta} + 2\lambda(1 + \psi(M_c)\hat{\rho})(\hat{u}_2 + \frac{u_2^*}{u_{02}})\frac{d\hat{u}_1}{d\eta} + \frac{d[\hat{R}_{12}(1 + \psi(M_c)\hat{\rho})]}{d\eta} = 0. \quad (15)$$

Achieving self-similar or fully developed conditions is very important for a detailed study of the flow, particularly when different shear layer cases with different configurations are compared. The shear layer starts to grow at different streamwise locations for different cases and, subsequently, the beginning of the self-similar zone is also very different. Mehta and Westphal 28 identified the self-similar zone by observing: a) linear growth rate of the shear layer, b) collapse of the scaled mean velocity, and c) Reynolds stress profiles with constant peak values. In the present study, we established the self-similar zone where we got the linear growth rate of shear layer thickness and momentum thickness, the collapse of mean streamwise velocity profiles, and constant peak values of Reynolds stress components in the self-similar coordinates [19, 32]. Given that, the mean and turbulent statistics do not concurrently meet self-similar conditions at the same streamwise location, we designated the self-similar zone where all these criteria were fulfilled.

V. Analytical Expression of Transverse Velocity

The behavior of mean transverse velocity is very different on the high-speed and low-speed sides. We will explain this asymmetric behavior through the integral form of the self-similar continuity equation. Integrating the equation (14) from $\eta = -\infty$ to $\eta = +\infty$ and applying the boundary conditions, the integral form of the equation becomes:

$$\hat{u}_{2,+\infty} - \hat{u}_{2,-\infty} = -\psi(M_c) \int_{-\infty}^{+\infty} \hat{\rho}\hat{u}_1 d\eta - \frac{\psi(M_c)}{2\lambda} \int_{-\infty}^{+\infty} \hat{\rho} d\eta + \int_{-\infty}^{+\infty} \eta \frac{d\hat{u}_1}{d\eta} d\eta + \phi. \quad (16)$$

Empirically, it is observed that $\hat{u}_1$ is antisymmetric (Error function) and $\hat{\rho}$ is symmetric (Gaussian function) (figure 2). Therefore, the 1st and 3rd integrals on the right-hand side of the equation (16) are close to 0 and the simplified integral form of the compressible self-similar continuity equation becomes:

$$\hat{u}_{2,+\infty} - \hat{u}_{2,-\infty} = -\frac{\psi(M_c)}{2\lambda} \int_{-\infty}^{+\infty} \hat{\rho} d\eta + \phi. \quad (17)$$

If we reduce equation (17) to the incompressible limit (M_c goes to 0, $\psi(M_c)$ goes to 0), the integral form of incompressible self-similar continuity equation becomes:

$$\hat{u}_{2,+\infty} - \hat{u}_{2,-\infty} = \phi. \quad (18)$$

Equation (18) is the similar form presented by Wei et al. 9 for the incompressible case. From equation (17), it is evident that for the compressible spatially evolving shear layer, the difference between $\hat{u}_{2,+\infty}$ and $\hat{u}_{2,-\infty}$ is primarily due to the density variation inside the shear layer and centerline shifting. However, for the incompressible case, since density variation inside the shear layer is negligible, the difference between $\hat{u}_{2,+\infty}$ and $\hat{u}_{2,-\infty}$ is due to the centerline shifting only.

Now, we will extend our self-similar analysis to find an approximate equation for the transverse velocity. The transverse velocity of a planar shear layer has received little attention in the existing literature, likely due to very difficult accurate experimental measurements. Wei et al. 9 derived an approximate equation for the incompressible planar mixing layer, but there is no such expression for the compressible version. The analysis presented here is an extension of Wei et al. 9 for the compressible shear layer. Following the same steps as Wei et al. 9, but using our compressible continuity and momentum equations, the fully compressible version of the scaled mean transverse velocity is

$$\begin{aligned} \hat{u}_2 = & \phi\hat{u}_1 + \frac{1}{1+\hat{\rho}\psi(M_c)} [\psi(M_c) \int_{-\infty}^{\eta} \eta \frac{d(\hat{\rho}\hat{u}_1)}{d\eta} d\eta + \frac{\psi(M_c)}{2\lambda} \int_{-\infty}^{\eta} \eta \frac{d\hat{\rho}}{d\eta} d\eta + \int_{-\infty}^{\eta} \eta \frac{d\hat{u}_1}{d\eta} d\eta] - \frac{1}{1+\hat{\rho}_0\psi(M_c)} [\psi(M_c) \int_{-\infty}^0 \eta \frac{d(\hat{\rho}\hat{u}_1)}{d\eta} d\eta \\ & + \frac{\psi(M_c)}{2\lambda} \int_{-\infty}^0 \eta \frac{d\hat{\rho}}{d\eta} d\eta + \int_{-\infty}^0 \eta \frac{d\hat{u}_1}{d\eta} d\eta] + [\frac{1}{1+\hat{\rho}\psi(M_c)} - \frac{1}{1+\hat{\rho}_0\psi(M_c)}] [\psi(M_c) \frac{\lambda-1}{2\lambda} \int_{-\infty}^{+\infty} \hat{\rho}\hat{u}_1 d\eta + \frac{\psi(M_c)}{4\lambda} \int_{-\infty}^{+\infty} \hat{\rho} d\eta \\ & + \frac{1-\lambda}{2\lambda} \int_{-\infty}^{+\infty} \eta \frac{d\hat{u}_1}{d\eta} d\eta - \psi(M_c) \int_{-\infty}^{+\infty} \hat{\rho}\hat{u}_1\hat{u}_1 d\eta + 2 \int_{-\infty}^{+\infty} \eta\hat{u}_1 \frac{d\hat{u}_1}{d\eta} d\eta + \phi \int_{-\infty}^{+\infty} \hat{u}_1 \frac{d\hat{u}_1}{d\eta} d\eta]. \end{aligned} \quad (19)$$

We have already mentioned that empirically $\hat{u}_1$ is an anti-symmetric function (well approximated by an error function), and $\hat{\rho}$ is a symmetric function (well approximated by a Gaussian). Then, applying the functional forms of

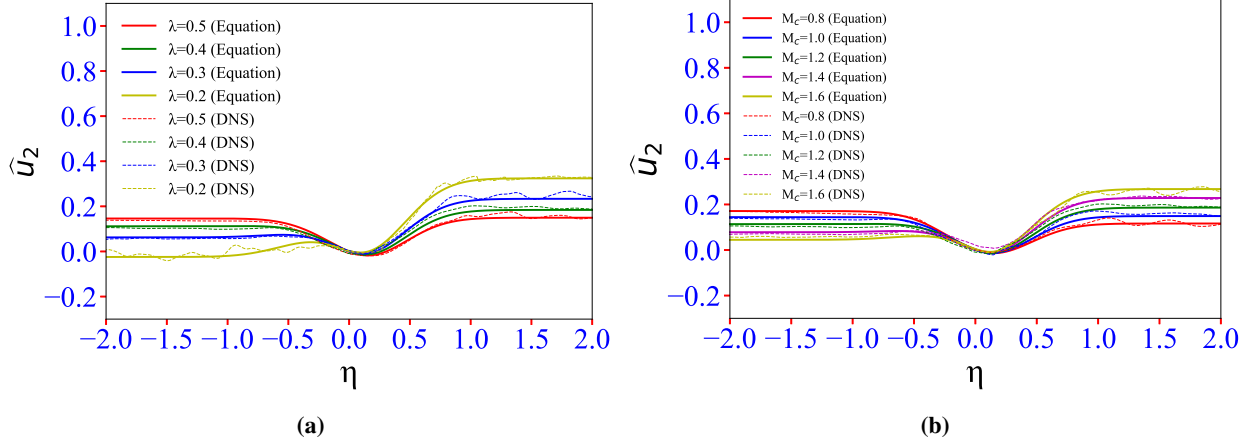


Fig. 3 The scaled mean transverse velocity profile in the self-similar coordinate compared with our approximate equation at different (a) velocity parameters and (b) convective Mach numbers. It is evident from the figure that the DNS results are reasonably close to the profiles obtained from our approximate equation.

$\hat{u}_1 = A \operatorname{erf}(B\eta)$, $\hat{\rho} = a e^{-B^2 \eta^2}$ in the integrals, the final analytical form of the approximate $\hat{u}_2$ equation is:

$$\begin{aligned} \hat{u}_2 = & \frac{\phi}{2} \operatorname{erf}(B\eta) - \frac{\eta e^{-B^2 \eta^2} \psi(M_c) [1 + \lambda \operatorname{erf}(B\eta)]}{2\lambda(1 - \psi(M_c) e^{-B^2 \eta^2})} + \frac{1}{1 - \psi(M_c)} \left[\frac{\psi(M_c) \sqrt{\pi} (1 + \lambda)}{8B\lambda} + \frac{1}{2\sqrt{\pi}B} \right] \\ & - \frac{1}{1 - \psi(M_c) e^{-B^2 \eta^2}} \left[\frac{\psi(M_c) \sqrt{\pi} [1 + \operatorname{erf}(B\eta)]}{8B} \left(\frac{1}{\lambda} + \frac{1}{2} [\operatorname{erf}(B\eta) - 1] \right) + \frac{e^{-B^2 \eta^2}}{2\sqrt{\pi}B} \right] \\ & + \left[\frac{1}{1 - \psi(M_c) e^{-B^2 \eta^2}} - \frac{1}{1 - \psi(M_c)} \right] \left[\frac{-\psi(M_c)(3 - \lambda)}{12\lambda B} + \frac{1}{\sqrt{2\pi}B} \right]. \end{aligned} \quad (20)$$

In equation (20), the values of A , B , and a were used as 0.5, 2.05, and -1.0, respectively, as suggested by the data. These values also significantly simplify the equation. Equation (20) represents the compressible form of the approximate self-similar scaled transverse velocity equation. Subsequently, this equation serves as the basis for investigating the characteristics of the scaled mean transverse velocity, and its outcomes are compared with current DNS findings across varying convective Mach numbers and velocity parameters. From figure 3, it is evident that the profiles from our approximate equation are reasonably close to the DNS results. In the incompressible limit, $\psi(M_c)$ is 0 and the incompressible form of the approximate self-similar transverse velocity equation becomes:

$$\hat{u}_2 = \frac{\phi}{2} \operatorname{erf}(B\eta) + \frac{1}{2\sqrt{\pi}B} - \frac{1}{2\sqrt{\pi}B} e^{-B^2 \eta^2}. \quad (21)$$

Equation 21 is similar to the scaled mean transverse velocity derived by Wei et al. 9.

VI. Entrainment

To understand the interactions between the free streams and the edges of the shear layer, entrainment analysis is very important [33]. Entrainment is the physical process by which the free-stream fluids are dragged into the turbulent zone within the shear layer [2, 33–35]. In the literature, entrainment has been divided into two categories (i) engulfment and (ii) nibbling. In engulfment, pockets of non-turbulent fluids are trapped in the turbulent zone whereas, in nibbling, free-stream fluids become part of the mixing layer due to viscous diffusion by small-scale eddies [2, 33].

As previously noted, in the constant density spatially developing shear layer, the centerline shifts towards the low-speed side, and due to this change of the orientation, the shear layer entrains varying amounts of fluid from each stream [2, 34]. The entrainment ratio (E_v) serves as an effective means to comprehend this particular phenomenon. When the shear layer exhibits self-similar growth, the volumetric entrainment from the two-speed sides can be calculated from the magnitude of the free stream vector and the angle of the edges of the shear layer [34]. The theoretical estimation of the entrainment ratio presented in this study is motivated by the analytical solution developed by Abramovich et al. 36 for the classical problem of the turbulent mixing layer of the two infinite streams. Following the entrainment analysis presented by D'Ovidio and Coats 34, and Abramovich et al. 36, the entrainment ratio can be expressed as:

λ	V_{01}	V_{02}	$V_{01}+V_{02}$	V_{01}/V_{02}
0.2	0.0273	0.0165	0.0438	1.65
0.3	0.0382	0.0215	0.0597	1.776
0.4	0.0572	0.0302	0.0874	1.896
0.5	0.0758	0.0382	0.114	1.98

Table 1 Variation of total entrainment and entrainment ratio with velocity parameter λ . The convective Mach number is 1.2 for all these cases.

M_c	V_{01}	V_{02}	$V_{01}+V_{02}$	V_{01}/V_{02}
0.8	0.0546	0.0369	0.0915	1.48
1.0	0.0551	0.0329	0.088	1.67
1.2	0.0572	0.0302	0.0874	1.896
1.4	0.0575	0.0263	0.0838	2.186
1.6	0.0578	0.0228	0.0806	2.535

Table 2 Variation of total entrainment and entrainment ratio with convective Mach number M_c . The velocity parameter is 0.4 for all these cases.

$$E_v = \frac{V_{01}}{V_{02}} = \frac{\hat{u}_{2h} + \frac{u_{2*}}{u_{02}} + \frac{1}{2}(\hat{u}_{1h} + \frac{1}{2\lambda})(1 + \phi)}{\hat{u}_{2l} + \frac{u_{2*}}{u_{02}} + \frac{1}{2}(\hat{u}_{1l} + \frac{1}{2\lambda})(1 - \phi)}, \quad (22)$$

where subscript h denotes the high-speed side, subscript l denotes the low-speed side and

$$V_{01} = \Delta U \frac{d\delta}{dx} [\hat{u}_{2h} + \frac{u_{2*}}{u_{02}}] + \frac{1}{2} \Delta U (\hat{u}_{1h} + \frac{1}{2\lambda}) \frac{d\delta}{dx} (1 + \phi), \quad (23)$$

$$V_{02} = \Delta U \frac{d\delta}{dx} [\hat{u}_{2l} + \frac{u_{2*}}{u_{02}}] + \frac{1}{2} \Delta U (\hat{u}_{1l} + \frac{1}{2\lambda}) \frac{d\delta}{dx} (1 - \phi). \quad (24)$$

By replacing $(\hat{u}_2 + \frac{u_{2*}}{u_{02}})$ in equation (23) and (24) from our $\hat{u}_2$ derivation, using functional form of $\hat{u}_1$ and $\hat{\rho}$ and assuming that at the edges of the shear layer $erf(B\eta) = \pm 1$, and $e^{-B^2\eta^2} = 0$ to avoid any arbitrariness in the choice of the edge of the shear layer, E_v for compressible shear layer becomes:

$$E_{v,comp} = \frac{\frac{4\lambda}{\sqrt{2\pi}B} + (1 + \lambda)(1 + 3\phi) + \frac{\psi(M_c)\sqrt{\pi}}{B} [1 + \frac{\lambda}{3}]}{\frac{4\lambda}{\sqrt{2\pi}B} + (1 - \lambda)(1 + \phi) + \frac{\psi(M_c)\sqrt{\pi}}{B} [-1 + \frac{\lambda}{3}]}. \quad (25)$$

In the incompressible limit, $\psi(M_c)$ is 0 and E_v for incompressible shear layer is:

$$E_{v,incomp} = \frac{\frac{4\lambda}{\sqrt{2\pi}B} + (1 + \lambda)(1 + 3\phi)}{\frac{4\lambda}{\sqrt{2\pi}B} + (1 - \lambda)(1 + \phi)}. \quad (26)$$

In the limit when λ goes to 0, ϕ goes to 0 and E_v becomes 1 for both incompressible and compressible shear layers. For compressible shear layer at $\lambda=0$, (by definition) M_c becomes 0 and therefore $\psi(M_c)$ becomes 0, which eventually makes $E_v=1$. $B=1.812$ is used by Wei et al. 9 for the incompressible shear layer based on the available data. Our analysis indicates that a value of $B=2.05$ fits our profiles effectively. Since the values B are reasonably close, $B=2.05$ is used in this analysis.

Equations (25) and (26) indicate that the incompressible E_v expression is a function of λ and ϕ , while the compressible E_v is influenced by λ , ϕ , and M_c . The value of ϕ depends on the convective Mach number (M_c), velocity ratio (r), and density ratio (s). Since the density ratio is kept unity in this paper, the variation of ϕ is dependent on

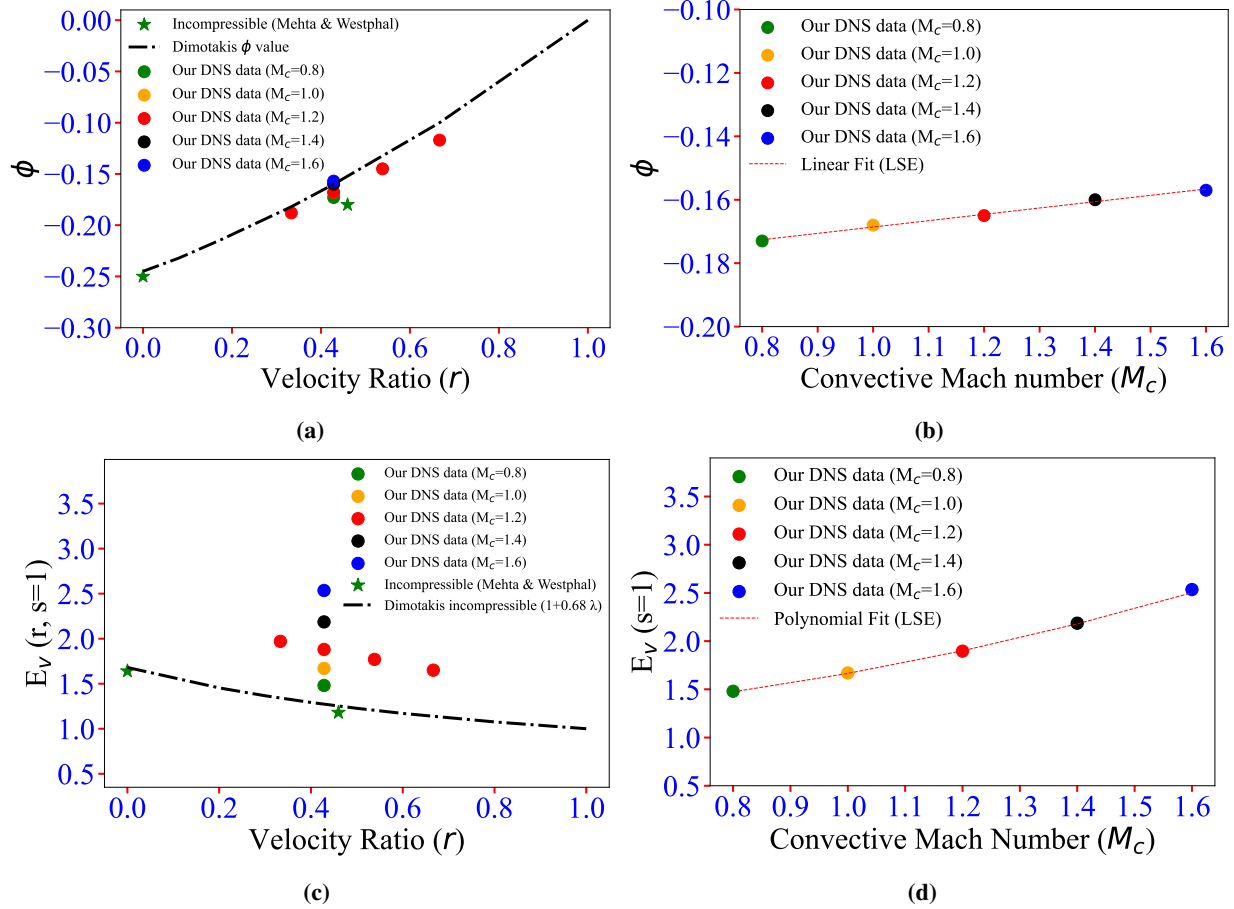


Fig. 4 (a) Incompressible and compressible values of ϕ at different velocity ratios and convective Mach numbers. The incompressible ϕ values are taken from Mehta and Westphal 28 while the compressible ϕ values are derived from current simulations. The dashed line is extracted from $E_v=1+0.68\lambda$ formula presented by Dimotakis 2 (b) Variation of ϕ with different convective Mach numbers. ϕ varies linearly with convective Mach number in the range of $M_c=0.8$ to 1.6. The equation of the linear fit using the Least Square Error (LSE) technique is $\phi=0.02M_c-0.1886$. However, it is not implied that a similar linear correlation would exist outside this M_c range and therefore, it is recommended not to use this linear fit outside this range of M_c . (c) Entrainment ratios calculated from the ϕ values presented in figure (a) using our incompressible and compressible E_v equations. Employing the incompressible ϕ values from Mehta and Westphal 28 in our derived incompressible E_v equation yields results that closely align with Dimotakis curve. (d) Variation of E_v with convective Mach numbers. A second-order polynomial ($E_v=0.55M_c^2-0.035M_c+1.15$) using the LSE technique is fitted to our E_v vs. M_c data. However, it is recommended not to use this formula outside this M_c range.

M_c and r . In the literature, the variation of ϕ with M_c has not been reported. Therefore, our DNS data is used to find the correlation between ϕ and M_c (Figure 4 b). In the range of higher M_c ($M_c=0.8-1.6$), a linear correlation is found between ϕ and M_c . However, it is not implied that a similar linear correlation would exist outside this range of M_c .

The relation between ϕ and velocity parameter (λ) or velocity ratio (r) is not straightforward. In the literature, the variation of ϕ with r has rarely been reported, especially at higher velocity ratios [9]. In this paper, we have reported the variation of ϕ with velocity ratio based on our current DNS data on the top of the incompressible ϕ values from previous studies [9, 28], as shown in figure 4 (a). At velocity ratio 1, ϕ approaches 0. Due to the limited number of data points available for ϕ across different velocity ratios, establishing a definitive relationship between ϕ and r is challenging.

In figure 4 (c), the compressible and incompressible E_v at different velocity ratios have been plotted and compared with $E_v=1+0.68\lambda$ formula suggested by Dimotakis 2. Employing the ϕ values from Mehta and Westphal 28 in our incompressible E_v equation yields results that closely align with Dimotakis 2 model. As the compressibility increases, the value of E_v increases which is evident from Figures 4 (c) and (d).

To further examine the variation of E_v with M_c and λ , the total entrainment and entrainment in each stream are investigated. Tables 1 and 2 show the variation of total entrainment, entrainment ratio, and entrainment in both streams at different velocity parameters and convective Mach numbers, respectively. Total entrainment and entrainment ratio increase with the increase of velocity parameters as shown in table 1. Dimotakis 2 and Brown 4 found a similar trend of increasing total entrainment and entrainment ratio with increasing velocity parameters. Entrainment in individual streams also increases with increasing velocity parameters. For a fixed density ratio 7, Brown 4 showed a similar trend of entrainment in each stream with velocity parameters. Table 2 shows that the total entrainment decreases while the entrainment ratio increases with increasing compressibility. Due to compressibility, the total entrainment decreases, which is connected to the reduced growth rate. The excess entrainment in the high-speed side with increasing compressibility is responsible for this higher entrainment ratio.

VII. Conclusions

Direct Numerical Simulations (DNS) of a spatially developing supersonic turbulent shear layer are conducted for a detailed study of self-similarity, transverse velocity, and entrainment. Fully compressible dimensionless mass, momentum, energy, scalar, and equation of state are solved with high-order numerical methods. The effects of convective Mach numbers and velocity parameters on transverse velocity and entrainment are studied, as these parameters appear naturally in our self-similar equations and for a global constant density shear layer; these are the two most important factors affecting the evolution, structure, and growth rate of the shear layer. To conduct a well-organized and systematic study, five different convective Mach numbers were chosen keeping the velocity parameter fixed and four different velocity parameters were chosen keeping the convective Mach number fixed. A wide range of grid sizes and domain sizes were considered in the convergence tests to determine the accuracy of the numerical results.

In the present investigation, we extend upon prior research by employing the fully compressible form of self-similar continuity, and streamwise momentum equations. For the first time, we incorporated compressibility, centerline shifting, and ϕ in the self-similar analysis. We also showed that the convective Mach number and velocity parameter emerge naturally in the self-similar equations, and therefore we have studied their effects on the transverse velocity and entrainment. The transverse velocity shows asymmetric behavior in the high-speed side and low-speed side at different convective Mach numbers and velocity parameters. The integral form of the self-similar continuity equation indicates that this asymmetric behavior of transverse velocity is due to the density variation in the core of the shear layer and the centerline shifting. Using the self-similar continuity and momentum equations, we derived an approximate equation for the transverse velocity and the profiles from the approximate equation are reasonably close to our DNS data at different convective Mach numbers and velocity parameters.

As expected, our results indicate that for a constant density shear layer, the centerline shifts towards the low-speed side, and hence ϕ is always negative. The absolute value of ϕ increases with increasing velocity parameter. Our simulation data also indicates that in the convective Mach number range ($M_c=0.8$ to 1.6) considered, ϕ shows a linear variation with M_c . This does not imply that outside this range of convective Mach numbers, ϕ would exhibit similar linear behavior. Furthermore, it is challenging to establish a definitive trend between ϕ and λ with a wide range of flow parameters.

In this study, we conducted an entrainment analysis to understand the interactions between the free streams and the edges of the shear layer. Using the approximate equation of transverse velocity, we derived a general expression for the entrainment ratio, which is the ratio of entrainment in high- and low-speed streams. Employing the ϕ values from the literature in our incompressible entrainment ratio equation yielded outcomes that were closely aligned with

those predicted by the model [2]. The entrainment ratio was found to increase with the convective Mach number or compressibility and high- and low-speed velocity difference or velocity parameter, favoring excess entrainment on the high-speed side. However, the total entrainment is found to decrease with increasing the compressibility and decreasing the velocity difference.

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